Multi-agent Visual Question Answering with Knowledge Base

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Abstract

In this study, we explore an innovative approach to Visual Question Answering (VQA) by incorporating a multi-agent framework that leverages both vision and large language models (VLMs and LLMs) along with an integrated knowledge base. Building upon recent advancements, we propose a setup that introduces CaptionBot, RetrieveBot and IntegrateBot to enhance VQA performance. However, unlike existing methods, our focus includes a knowledge base accessed by an additional agent, aiming to improve both the accuracy and explainability of the answers generated. We investigate the impact of various prompts on performance and provide both logical and experimental evidence supporting the multi-agent setup as a promising approach for this task.

1 Introduction

Visual Question Answering (VQA) is a complex task in machine learning which is highly relevant for advancing multimodal AI capabilities. It involves generating accurate and contextually appropriate answers and thereby pushing the boundaries of how machines comprehend and interact with visual and textual information. Recent vision-language models (VLMs) pre-trained on huge image-text data have made notable strides in addressing VQA tasks. A crucial aspect of these models' effectiveness in VQA tasks is their ability to leverage their pre-trained foundational knowledge through prompting, without the need for task-specific fine-tuning. This involves using carefully designed prompts in zero-shot settings to including examples of triplets [image, question, answer] in few-shot settings (Li et al., 2023), (Alayrac et al., 2022). The significance of selecting appropriate prompts was also investigated and demonstrated in (Awal et al., 2024). Naturally, the optimal prompts differ for each model.

The task of Knowledge-Base Visual Question Answering (KB-VQA) has been approached through various methodologies over time. This task involves answering questions based on an image, requiring not only visual information but also additional knowledge, which could be either common knowledge or information stored in a database. Initial approaches utilized convolutional neural networks and recurrent neural networks to map input images and questions to answers. These methods operated under the assumption of a finite set of possible answers, thus not supporting open-ended answer generation as described in (Wang et al.). In contrast, recently developed large language models (LLMs) possess embedded knowledge and reasoning capabilities, enabling them to answer knowledge-based questions without additional retrieval from a knowledge base. However, in tasks where questions are derived from an accessible knowledge base, it is reasonable to hypothesize that the complexity of the required reasoning may benefit from explicitly using this knowledge base. Recent solutions have proposed combining the implicit knowledge of LLMs with explicit knowledge bases to enhance performance (Gui et al., 2022).

Having access to LLMs specialized in reasoning and VLMs specialized in image analysis, it remains crucial to study how to integrate these models. The goal is to enable them to perform human-like reasoning based on an image, a question, common sense knowledge, and specialized knowledge.

Building on mentioned advancements in the field, we have designed a setup that leverages the strengths of each component agent. Specifically, we employ a vision-language model (VLM) to generate captions and address simpler questions. Given that VLMs are less specialized in reasoning compared to large language models (LLMs),

integrating an LLM enhances the system's reasoning capabilities. Our study examines the impact of incorporating a knowledge base on VQA performance, focusing on how pre-trained LLMs adapt when supplemented with pertinent information retrieved by an agent dedicated to identifying the top-k useful facts. Additionally, we explore the dynamics of multi-agent communication, allowing LLMs to interact with VLMs by posing supplementary questions to obtain more detailed information. The **main contribution** of our work lies in the multi-agent framework, where agents determine the additional information they need and communicate with other agents to obtain it.

2 Related Work

Visual Language Transformers. In recent years, multimodal transformers have been developed that process both text and image components as input. These models have demonstrated impressive performance on various vision-language tasks, including image captioning, masked word prediction, and Visual Question Answering (Fields and Kennington, 2023). However, they do not incorporate external knowledge, highlighting the need for solutions that integrate Visual Language Transformers with external knowledge sources, as discussed in (Gui et al., 2022). Notable examples of Visual Language Transformers (VLTs) include BLIP-2 (Li et al., 2023), which we employ as an agent, and Flamingo (Alayrac et al., 2022).

LLM based methods To utilize large language models (LLMs) in a zero-shot manner, the prompt engineering is necessary (Guo et al., 2023) (Awal et al., 2024). In few-shot scenarios, we provide the model with examples to guide its responses (Yang et al., 2022). In this approach, the language model receives an image description as a prompt and generates an answer, effectively using the LLM as an unstructured and implicit knowledge base. Another method involves using the LLM to generate a question-aware caption from the VLM, as demonstrated in PromptCap (Hu et al., 2023), to enhance the quality of the input for the LLM.

Multi-agent approach. The conceptual goal of multi-agent frameworks is to emulate human reasoning by simulating different steps or abilities through various agents. For instance, in the paper (Li et al., 2024), multiple agents generate outputs, which are then aggregated through voting to determine the final answer for the LLM task. Additionally, several papers have proposed a "debate" mechanism among LLM agents to reach a final answer (Du et al., 2023). This multi-agent approach can be particularly beneficial for Visual Question Answering (VQA) tasks, as it facilitates the integration of multiple modalities. It enables agents not only to exchange raw information but also to derive deeper insights through inter-agent discussions. For example, the paper (Wang et al., 2023d) describes a Seeker agent that asks additional questions to a Responder agent, while another agent reasons about all the gathered information. However, this paper does not utilize external knowledge, addressing tasks that may require such information without having access to it.

3 Model methodology

In this section, we describe the setup and functionalities of each agent as well as the procedure by which they communicate. This setup aims to leverage the high capabilities of the VLM for image description, the reasoning abilities of the LLM, and the retrieval skills of RetrieveBot for finding hints. Well-prepared interaction will allow for the optimal use of their best skills.

3.1 The CaptionBot

The VLM model excels in describing images and answering questions about their content. Due to this specialization, its reasoning capabilities are limited. While it can handle VQA tasks, it often makes mistakes with questions requiring broader knowledge or reasoning. For example, being asked a question "What object in this image can be male or female?" it answers "a tennis ball", clearly lacking the ability to perform deeper reasoning. To enhance its capabilities, we use it as an agent - CaptionBot, capable of answering questions and describing images. CaptionBot is a pre-trained model specialized in generating image captions and providing answers to short questions. Initially, CaptionBot prepares an image description, which other agents use to begin their reasoning process. Later in the algorithm, CaptionBot answers additional questions that may help gather more detailed information about the image. To describe this procedure, consider the following example: when asked, "Which object in this image is large?" the caption "the wing of a plane" is insufficient if the ques-

Algorithm

```
procedure multi_agent(prompt_templates, question, image)
    Initialize state(image, question)
    state["caption"] = CaptionBot.caption()
    state["hints"] = RetrieveBot()
    state["supportive question"] = InterprateBot.generate()
    state["caption2"] = CaptionBot.answer()
    return IntegrateBot.answer()
```

Algorithm 1: The procedure of agent communication. All the information is stored in a "state" dictionary

tion pertains to the mountains in the background. Therefore, we introduce an additional question to CaptionBot, posed by IntegrateBot, to obtain more relevant information about the image.

3.2 The RetrieveBot

The task of RetrieveBot is to search the knowledge base for the top-k most relevant facts related to a given question and image description. By identifying connections that may be crucial for answering the question, RetrieveBot often provides valuable information that enables IntegrateBot to perform accurate reasoning.

3.3 The IntegrateBot

IntegrateBot is a pretrained Large Language Model (LLM) that we interact with using prompts. Due to its high generative and reasoning capabilities, we use it to manage other agents and integrate the overall context to provide the final answer. Throughout the algorithm, it performs two tasks:

1. Basing on the context consisted of the question, the caption to the image and hints from the RetrieveBot, it formulates an additional question to the CaptionBot.
2. Basing on all the context from agents answers the question.

3.4 Interaction Algorithm

To fully leverage the reasoning capabilities of IntegrateBot, we sought to provide it with the highest quality image descriptions and proposed facts from the Knowledge Base when answering questions. Inspired by the paper (Wang et al., 2023d), we employed the mechanism of asking additional questions to CaptionBot. Initially, CaptionBot generates a caption based solely on the image. Subsequently, RetrieveBot searches for hints not only based on the question but also on the image description provided by CaptionBot. Based on the data obtained so far, IntegrateBot formulates an additional question, which CaptionBot answers using only the additional question and the image. By passing the entire context to IntegrateBot, we generate the final answer.

4 Experimental Set-up

4.1 Dataset.

The FVQA v2 (Lin et al., 2023) dataset is a comprehensive resource designed for the Fact-based Visual Question Answering task. It extends traditional VQA by incorporating external factual knowledge, enabling models to answer questions that require information beyond the image content. FVQA v2 includes over 8,000 questions and images with associated factual triples derived from knowledge bases, ensuring a rich context for complex question-answer pairs. We selected the FVQA v2 dataset for our VQA experiment due to its emphasis on integrating visual content with factual reasoning and because of the knowledge base that is directly useful in reasoning. The dataset's diverse and challenging nature provides a robust benchmark for evaluating the effectiveness of our approach in handling real-world scenarios where visual understanding and factual knowledge must be synergistically applied.

We combined the training and test datasets from the original source. Firstly, a large training set was unnecessary due to the zero-shot-like design of our model. Secondly, we aimed to include adversarial samples introduced by the FVQA v2 paper in the test set to more thoroughly evaluate the model's reasoning capabilities. However, we set aside approximately 1% of the samples to create a small working dataset used for prompt engineering and the development of the communication algorithm. This approach resulted in a test set comprising over 8,000 questions for evaluating our model.

Knowledge Base. The Knowledge Base comprises over 225,434 short sentences corresponding to fact triplets (e.g., "[car] has [4 wheels]"). Each question in the dataset is associated with one of these triplets. Although the triplets primarily represent common-sense knowledge, the extensive size of this Knowledge Base is substantial compared to the number of questions.

4.2 Agent's Models

BLIP-2. We use BLIP-2 model for obtaining answers to questions and for obtaining image descriptions, we used prompts recommended by the model's authors. This open source, pretrained model has robust performance in natural language understanding tasks and is integral to our experimental setup. Its ability to generate accurate captions and integrate contextual information enhances the accuracy of our proposed approach. Leveraging its capabilities, we aim to achieve nuanced insights and results in our study.

Llama. We employed an easily accessible open-source text-to-text model, which is freely available for download and can be directly executed on our device. This model is relatively compact and specifically designed to operate within our RAM limitations. Despite its size—less than 10 billion parameters—the model demonstrates surprisingly robust performance. Within our framework, Llama is tasked with step-by-step reasoning, involving additional questioning and delivering the final answer.

Sentence-BERT. We utilize the Sentence-BERT model to encode sentences or paragraphs into embedding vectors, as described in https://arxiv.org/abs/1908.10084. Despite being introduced five years ago, it continues to deliver high-quality performance in search tasks, functioning as a Retrieval Agent. The architectural design of BERT renders it unsuitable for semantic similarity search. Within our framework, Sentence-BERT is tasked with querying the Knowledge Base to identify relevant facts by measuring the similarity between these facts and the concatenated question and caption.

4.3 Metrics.

We examined the exact match between the algorithm-generated answers and the ground truth.

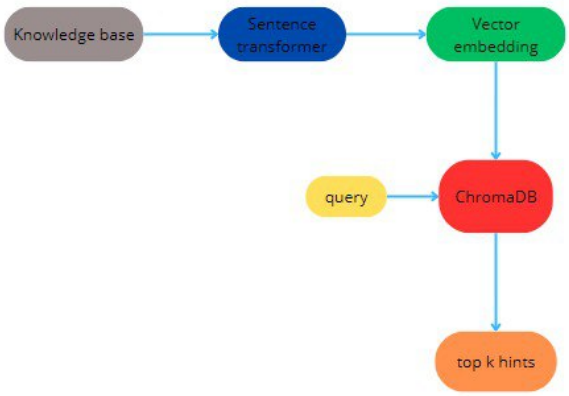


Figure 1: RetrieveBot algorithm diagram

Due to recurring differences such as the use of articles (a, an, the), punctuation at the end of answers, and variations in capitalization at the beginning of answers, we performed simple normalization on these aspects before comparison.

4.4 LLM Prompt engineering

Quality of a prompt might significantly the quality of an output (Sahoo et al., 2024). During the prompt engineering process we've heavily relied on this guideline (llama.meta), incorporating to our prompt Role Based prompt engineering technique, as it increases both relevance of the answer and its accuracy. We've also decided to go with Zero-Shot Prompting, explicitly instructing the model to return short outputs.

The final prompts 3 were created via trial and error search for prompts that would result in the most satisfying results. This whole process has been performed on a special small subset of our dataset excluded from any further tests.
An example of one of the problems that we've stumbled upon during our initial tests: during our first attempt we didn't include an explicit request for the integrator's response to be short. This resulted in way too long responses and thus the second part of the prompt (as in 3) has been chosen from a bunch of statements with similar meaning, but different phrasing and the most promising one has been added in the end to the final version of the prompt.

Another example: while creating a prompt for the question generating model we have noticed that it sometimes is too enthusiastic and outputs multiple questions despite being prompted to return

VLM	caption+RAG+LLM	XXLLM (ours)
36.77%	40.52%	46.96%

Table 1: Performance scores of out model in comparison to versions with poorer structure.

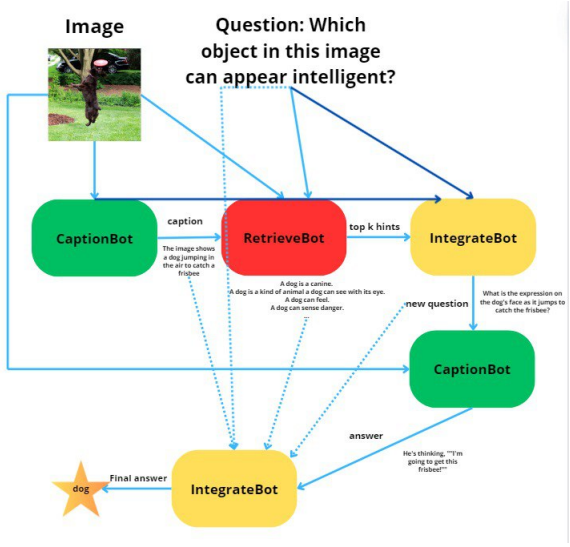


Figure 2: A graph illustrating our pipeline.

only a single one. Thus we've rephrased wording so that it contains an emphasis on the fact that we want only one question and not more 3, which turned out to in fact successfully mitigate this issue.

4.5 Model Analysis and Ablations

We conduct a series of ablation studies to analyze our model and identify the sources of improvement, with a particular focus on the complexity of the reasoning setup using accessible tools and models. Our goal is to identify and understand the effective components of our method and the reasons behind their success.

We evaluated the performance of the zero-shot Vision-Language Model (VLM) using the same model as CaptionBot, BLIP-2. Subsequently, we conducted a *caption+RAG+LLM* experiment, omitting only the part involving supplementary questions. Finally, we compared these versions to the final setup described above. Additionally, we conducted detailed case analyses on multiple questions to better understand the advantages and limitations of each approach.

5 Results and Discussion

5.1 Numbers

We observed significant performance differences among the versions listed, empirically validating our predictions regarding architectural improvements. We evaluated scores exclusively on baselines we constructed for two main reasons. Firstly, due to limited computational resources, we conducted our experiments on open-source machines, making direct comparison with state-of-the-art solutions based on the best available models challenging. Secondly, we encountered issues with the documentation of the FVQA v2 dataset. Despite considerable effort, our team was unable to locate the labeling of samples and their division into the specified train and test subsets as described in the paper. Nonetheless, we proceeded with experiments using the selected dataset and obtained interpretable results.

5.2 Analysis of differences

In addition to measuring the accuracy of each model, we conducted an analysis of the responses generated by each of the constructed models. Two sample questions along with their respective answers are presented in Figures 3 and 4. Figure 3 features a question requiring basic knowledge, "Which object in this image desires a haircut?" We observe that the Simple VLM answer provides an incorrect response without engaging in reasoning ("A tennis racket"), selecting one of the objects in the image inaccurately. In contrast, the Simple RAG answer (from the second experiment) approaches the correct answer by leveraging the LLM and Knowledge Base ("None, the man has hair"). However, the LLM's additional reasoning step imposed through interaction with the VLM enhances the response - "The person". Based on RAG retrievals, we observe the quality of information retrieved by RetrieveBot. This example underscores the advantage of our designed model over less sophisticated approaches.

Several challenges arose during our testing phase with our approach. One notable issue was that although we observed better performance and question answering during the re-questioning the CaptionBot, there was still unsolved problems with inaccurately described image. For instance, in the question "Which object in this image is a public transportation device?", Caption-

Bot failed to include a bus in its description, which subsequently affected the IntegrateBot's ability to pose appropriate questions in this context.

5.3 RAG

Often, the questions in the dataset refer to non-obvious relationships between objects. By employing RAG, the system could quickly and accurately search for the most relevant hints to help answer the questions. One of the primary advantages of this approach are that the response, containing, for example, the three most relevant results, is very concise. Another one is that queries executed in this way are much faster than those answered by an LLM. In other words, we can efficiently obtain several potential, independent hints, which, in contrast to prompting an LLM, would require separate calls or cause dependencies during response generation. Additionally, RAG has the advantage over reasoning with LLMs in that it provides more predictable answers and does not hallucinate.

5.4 Comparing answers.

We observed several issues with the provided answers other than we fixed:

1. Ambiguity in whether the answer should be singular or plural in both the ground truth and the model's responses.
2. Correct answers that included additional, accurate words.
3. Synonymous answers.

We believe it would be beneficial to test other metrics for evaluating answer accuracy, as there are metrics based on word embeddings or evaluation using LLMs available. It may be the object of future work.

6 Conclusions

Multi-agent system plays a crucial role because it allows us to conduct more complex reasoning processes. The knowledge base provides ready-made connections between the content of the question and objects in the image. This often gives other agents a solid foundation for reasoning, asking for details, or strongly indicating a candidate for the answer.

However, multi-agent systems have their limitations – one can't use too many agents at once, as they tend to deviate from the main question and the actual content of the image.

Choosing prompts to standardize communication between agents is very important. It's essential to clearly signal how long the response should be, what it should contain, and even what it should not contain.

Overall the improvement in performance observed when retrieving facts from the Knowledge Base suggests that this approach is beneficial even for Large Language Models (LLMs) believed to already contain common-sense knowledge. This indicates that the setup helps to strengthen the reasoning capabilities of LLMs.

6.1 Code availability

The code and model outputs are available here.

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```
llm_image_question_role = """
    You are an AI assistant designed to ask a new question about an image.
    You are given an original question refers to the image and you are provided with the image caption.
    Your answer should contain exactly one follow-up question about the image.
    Don't answer the main question. Don't write anything else apart from the follow-up question.
    """
llm_integrator_role = """
    You are an AI assistant designed to answer simple questions based on the image caption.
    Your answers should be as short as possible, referring to a single thing.
    """
llm_integrator_prompt_template = 'Context: {context} Question: {orig_question} Answer:'
```

Figure 3: Exact prompts used in our experiments

question: Which object in this image desires haircut?
target answer: a person
simple vlm answer: A tennis racket
simple rag answer: None, the man has hair.
final arch answer: The person.
simple rag retrieval: a hair is for decoration. hair is cut there. Hair is on top of the head. person wants hair. this man has hair
final arch context: vlm_caption: The image shows a man holding a tennis racket and a ball on a tennis court.
rag_retrievals: person wants hair. a person wants a haircut. person wants a nice haircut
vlm_question: What is the person's hairstyle in the image?
vlm_answer: A bandana

Figure 4: An example where only the final model gave the correct answer

question: What can you see in this image is related to strike?
target answer: tennis racket
simple vlm answer: The man is playing tennis
simple rag answer: Tennis racket.
final arch answer: A tennis racket.
simple rag retrieval: tennis racket is related to strike. racquet is related to hit. racket is related to hitting ball. racket is related to hitting. A tennis ball is hit by a tennis racket
final arch context: vlm_caption: The image shows a man holding a tennis racket and a ball on a tennis court.
rag_retrievals: tennis racket is related to strike. racquet is related to hit. A tennis ball is hit by a tennis racket
vlm_question: What is the object being held by the man in the image that is related to strike?
vlm_answer: A tennis ball

Figure 5: An example where the second model with retrieval and the final model gave the correct answer